THEORETICAL RESEARCH PREPRINT

Counterdiabatic Sheaf Transport on Finslerian Chromatin Manifolds

An Identity-Constrained, Energy-Aware Framework for Partial Cellular Rejuvenation

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Research status: Conceptual and methods framework with a reproducible synthetic demonstration. No biological cohort was downloaded or analyzed for this manuscript; no wet-lab experiments were performed. All numerical results are generated from explicitly constructed toy dynamics and must not be interpreted as biological validation.

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Abstract

Cellular rejuvenation is a constrained state-transition problem: age-associated molecular signatures may be shifted toward younger reference states, yet a useful intervention must preserve lineage-specific function, avoid malignant or dysplastic transitions, remain reachable with available perturbations, and not assume that transcriptomic similarity is equivalent to organismal rejuvenation. We propose *Counterdiabatic Sheaf Transport* (CST), a mathematical and computational framework for testing whether geometry-aware, identity-constrained control improves this trade-off over unconstrained state matching. The framework combines four ideas: (i) an information-geometric representation of multimodal cell states; (ii) an asymmetric Randers-Finsler path cost whose directional term is estimated from data rather than equated with entropy production; (iii) a cellular-sheaf consistency residual that operationalizes, but does not claim to prove, preservation of regulatory coherence; and (iv) a port-Hamiltonian-inspired dynamical model coupled to bounded, reachability-aware control. We derive the mathematical conditions under which each component has its intended interpretation and identify several tempting but invalid inferences: sheaf cohomology is not a certificate against cancer, a learned Hamiltonian is not automatically thermodynamic energy, and an inverse control formula is not an exact controller unless the requested velocity lies in the actuation range. A synthetic four-state dynamical example illustrates a generic trade-off: an aggressive intervention reaches a lower abstract age coordinate while causing larger identity-deviation and risk-proxy excursions; a constrained pulsed policy gives up some apparent endpoint improvement while remaining closer to illustrative guardrails. A 300-replicate parameter/noise stress test reproduces this qualitative trade-off in the chosen toy model. These results are computational demonstrations of the code and assumptions, not evidence about real cells. We specify a falsifiable empirical programme using existing single-cell aging atlases, time-resolved reprogramming datasets, chromatin-contact assays, and functional endpoints. The framework should be rejected or simplified if directional geometry, regulatory-consistency penalties, or structured control fail to improve held-out predictions or experimentally measured identity/function at matched rejuvenation. Its proposed contribution is thus an auditable integration and validation design, not a claim that rejuvenation, Finsler geometry, sheaf theory, or counterdiabatic control is individually new.

Keywords: partial cellular reprogramming; epigenetic aging; information geometry; Randers-Finsler metric; cellular sheaves; Hodge Laplacian; port-Hamiltonian neural ODE; constrained control; lineage identity; reproducible computational biology.

Authorship, accountability, and scope

Devanik Debnath originated the research direction supplied for this manuscript and serves as the human corresponding author. Xylia contributed AI-assisted literature synthesis, formalization, organization of the model, drafting, and development of the synthetic verification workflow. The designation of Xylia as a co-author is made at the human author's explicit request and is accompanied by transparent disclosure that Xylia is an AI system and cannot independently assume legal or scholarly accountability. Devanik Debnath retains responsibility for checking the literature, equations, code, citations, interpretations, and any future submission. The authors do not claim that the presented framework has been experimentally validated, clinically evaluated, or shown to reverse organismal aging.

The manuscript uses the terms *framework*, *hypothesis*, and *candidate method* deliberately. The strong version of the proposal—that a topological invariant can make cancerous dedifferentiation mathematically impossible, or that mitochondrial phase coherence is the single foundational cause

of cellular aging—is not supported by current evidence and is not adopted here. Instead, identity constraints are treated as measurable safety surrogates that require independent functional validation.

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1 Introduction: rejuvenation as a constrained transition

Aging is not a single molecular variable. It involves interacting processes including genomic instability, epigenetic alterations, loss of proteostasis, mitochondrial dysfunction, cellular senescence, altered intercellular communication, stem-cell exhaustion, and other changes summarized in contemporary accounts of the hallmarks of aging [1]. These processes differ across cell types, tissues, and individuals. A method that optimizes one molecular signature can fail to improve tissue function, and a method that increases plasticity can compromise the stable identity on which tissue architecture depends.

Partial reprogramming has established that some aging-associated molecular features can be altered by transient expression of reprogramming factors in selected experimental settings. Ocampo and colleagues reported amelioration of age-associated hallmarks in a premature-aging mouse model and related effects in older wild-type mice [3]. Lu and colleagues reported restoration of youthful methylation and transcriptomic features together with improved regeneration and vision-related outcomes in mouse models using OSK-based reprogramming [4]. Browder and colleagues found that the duration of in-vivo partial-reprogramming regimens influenced outcomes in physiologically aged mice [5]. Such studies justify asking how trajectories can be made controllable and measurable. They do not show that there is one universally safe rejuvenation trajectory, nor that all age-associated information can or should be reset.

The present proposal reframes the computational problem. Let a cell's latent molecular state be $x(t)$ and let $\mathcal{Y}_{\text{young}}$ denote a prespecified distribution of reference states for the relevant lineage, assay, sex, age window, and tissue context. A naive objective minimizes a distance to this target. A more responsible objective must also penalize loss of lineage markers, regulatory inconsistency, excessive control effort, dysplastic or oncogenic signatures, functional decline, and uncertainty. In practice, these requirements are partially observed and cannot all be compressed into one epigenetic-clock score.

We propose *Counterdiabatic Sheaf Transport* (CST) as an integrative model architecture, not a biological law. In CST, a path is evaluated with a direction-sensitive geometry; regulatory agreement is represented through a cellular sheaf; dynamics are modelled with an energy-accounting state-space architecture; and an auxiliary controller attempts to track a target path subject to actuation and safety constraints. Each component is tested against simpler alternatives so that complexity can be removed when it fails to help.

1.1 Research questions

The paper is organized around five questions.

1. Can direction-dependent path geometry improve the prediction of observed cell-state transitions beyond Euclidean or Riemannian baselines?
2. Can a sheaf-based regulatory-consistency penalty add information about lineage preservation beyond marker-gene and functional assays?
3. Can a port-Hamiltonian-inspired model improve model calibration or control robustness relative to an unconstrained neural ODE or a conventional state-space model?
4. Under what controllability conditions can a counterdiabatic-style auxiliary input track a desired trajectory, and when must the framework explicitly declare the target unreachable?
5. Do these components improve held-out rejuvenation-versus-identity trade-offs in biological data, rather than only in a synthetic model constructed to exhibit them?

1.2 Contributions and claims ledger

The contributions are methodological and conditional.

- **A typed latent-state model.** We distinguish observed assays, latent molecular state, lineage identity, age-associated endpoints, and safety endpoints rather than identifying them with one another.
- **A defensible directional cost.** We specify a Randers-Finsler metric with a data-estimated directional covector and the regularity condition required for a valid Randers norm.
- **An operational use of sheaf consistency.** We use the coboundary residual and harmonics as structural descriptors, with independent lineage markers and functional assays as non-negotiable validators.
- **A reachability-aware control formulation.** We replace an unconditional pseudoinverse formula with a constrained least-squares controller and explicit feasibility tests.
- **A reproducible synthetic falsification demonstration.** Public code tests the stated toy model and quantifies the trade-off created by chosen assumptions; no biological conclusions are inferred from it.

Table 1: Claim classification used throughout the manuscript.

Class	Meaning in this paper
Established background	Results already supported in cited literature, such as the existence of cell-type-specific aging signatures and selected partial-reprogramming effects.
Mathematical identity	A claim that follows from explicit algebraic assumptions, such as $F(v) - F(-v) = 2\beta(v)$ for a Randers norm.
Modeling choice	A proposed representation selected for usefulness, not a discovered fact, such as a learned phase-decoherence coordinate.
Empirical hypothesis	A prediction that must be tested on independent biological observations.
Synthetic demonstration	Behavior observed only in a constructed simulation with chosen parameters.
Unsupported stronger claim	A proposition explicitly rejected as unproven, such as cohomology invariance being sufficient to prevent cancer.

2 Prior art and the boundary of novelty

A serious framework must acknowledge the mature traditions it borrows from. This paper is built by connecting these traditions in a testable way; it does not claim to have invented them.

2.1 Epigenetic aging and partial reprogramming

DNA methylation clocks and cell-type-specific aging signatures have made it possible to quantify some age-associated molecular changes, but such measures are not complete descriptions of biological age or function. The Tabula Muris Senis consortium generated a single-cell transcriptomic atlas across the lifespan of mice covering 23 tissues, providing a foundational resource for cross-tissue

aging analyses [2]. The atlas also illustrates a key modeling constraint: cellular composition and within-cell expression can both change with age. Analyses that do not separate these effects can confuse shifts in cell population proportions with within-lineage state transitions.

Partial cellular reprogramming is likewise prior art. Ocampo et al., Lu et al., and Browder et al. report encouraging but context-bounded results in mice [3, 4, 5]. Reviews emphasize both the potential of partial reprogramming and remaining questions about mechanism, durability, and safety [6, 7]. CST does not frame OSKM as a uniformly harmful sledgehammer: outcomes depend on factors, delivery, duration, tissue, and biological context. The framework instead asks how to quantify a target state change while separately observing lineage and safety.

2.2 Chromatin organization and phase separation

The idea that chromatin organization is influenced by condensate-like phase behavior is supported by experiments on HP1 and heterochromatin domain formation [8, 9]. Reviews discuss roles of phase separation and the inheritance of repressive chromatin states [10, 11]. However, the broad analogy between age-related chromatin change and spinodal coarsening is not sufficient to prove that a particular thermodynamic phase-separation process drives aging in vivo. CST therefore treats chromatin condensation, domain boundaries, and associated features as potential observed coordinates, not as a universal mechanism. A claimed phase transition would require measured order parameters, finite-size and noise analyses, hysteresis or scaling signatures, and independent replication.

2.3 Mitochondria, metabolism, and nuclear state

Mitochondrial metabolites and stress signals participate in mitochondria-to-nucleus communication and can influence epigenetic regulation [12]. Circadian NAD^+ metabolism has also been linked to chromatin remodeling [13]. These data motivate measurement of metabolic context and temporal variation. They do not establish the stronger chain proposed in the original concept: mitochondrial membrane-potential and NAD^+/NADH rhythms phase-lock all chromatin organization, their decoherence causes spinodal coarsening, and this is the primary arrow of biological aging. CST treats that causal chain as a specific hypothesis to test against alternatives, not a premise to assume.

2.4 Information geometry and optimal transport

Information geometry supplies metrics on statistical models, including Fisher information metrics. In cell biology, single-cell state transitions have been analyzed using optimal transport and inferred temporal trajectories [14], and RNA velocity estimates short-term transcriptional dynamics from spliced and unspliced RNA under model assumptions [15, 16]. Importantly, a recent 2026 study introduced SCIM, using Fisher information geometry and RNA velocity to quantify cell-phenotype transition manifolds [17]. Thus, using information geometry for cell-state trajectories is not new. CST's question is whether an explicitly asymmetric path cost, tied to a separately measured drift field and combined with structural and safety constraints, adds predictive or control value beyond these precedents.

2.5 Cellular sheaves and cohomology

Cellular sheaves have been developed as a way to represent local data and compatibility conditions across a cell complex. Hansen and Ghrist provide a spectral treatment relating sheaf Laplacians to

cohomology [18]. In this paper, a sheaf assigns vector spaces to regulatory loci and complexes and restriction maps that encode a selected abstraction of consistency across regulatory relations. This is a new proposed application within CST, but no priority claim is made that it is the first possible use of a sheaf in gene regulation, and no claim is made that cohomology constitutes a sufficient representation of biological identity.

2.6 Counterdiabatic control and port-Hamiltonian learning

Counterdiabatic driving originated in quantum control as a method for tracking prescribed state changes while suppressing unwanted transitions, and approximate protocols can be derived under constraints on available control terms [21]. Exact tracking is not generally available under arbitrary actuation constraints. Port-Hamiltonian systems separate interconnection structure, dissipation, and input ports; learned port-Hamiltonian neural networks have been used to model time-dependent dynamical systems [22, 23]. CST transfers these modeling principles into a proposed cell-state control problem. Such an analogy is mathematical, not evidence that a cell literally obeys a quantum adiabatic theorem or that the learned Hamiltonian is physical free energy.

2.7 Novelty statement

The defensible novelty claim is a *specific, falsifiable integration proposal*: an asymmetric state-space path cost, sheaf-based regulatory-consistency diagnostics, port-Hamiltonian-inspired dynamics, reachability-aware counterdiabatic-style control, and independent identity/function/safety tests are evaluated as one framework. A scoped literature search through 14 October 2026 found close prior work on every component and direct precedent for information-geometric cell manifolds. We did not perform a formal systematic review of all publications, patent filings, or unpublished work. Absolute novelty and priority are therefore unverified. The core result of this paper is the explicit formulation and validation plan rather than a claim of experimental discovery.

3 Problem formulation and assumptions

Let c index a cell type or lineage, d a donor or biological replicate, and t a measured time or age. The desired state is not an arbitrary low-age point in latent space; it is a reference distribution $\mu_{\text{young},c}$ conditional on lineage and assay. The state at time t is latent and only partially observed. Define

$$Y_t = \mathcal{H}(x_t, z_t, q_t; \eta) + \epsilon_t, \tag{1}$$

where x_t is the molecular state, z_t denotes external and metabolic context, q_t denotes functional phenotype, η denotes platform and batch effects, and ϵ_t denotes measurement noise. Y_t may include RNA counts, chromatin accessibility, chromatin contacts, histone/DNA modifications, metabolite measurements, imaging features, and functional readouts. These modalities have different noise models and cannot be treated as interchangeable coordinates without a validated integration method.

We use a low-dimensional latent representation $x_t \in \mathcal{M} \subseteq \mathbb{R}^d$ only as an approximation to the relevant state manifold. The latent chart may be local and can fail across poorly sampled transitions. A coordinate transformation should not change a properly defined geometric result; neural embeddings and dimension reduction, in contrast, can distort distances and orientation unless checked. The target distribution, features, normalization, and state-estimation procedure must be frozen before held-out evaluation.

3.1 Assumption ledger

1. **Measurement sufficiency (A1).** The measured features resolve the molecular and functional dimensions material to the tested lineage and intervention. This assumption is unlikely to hold universally; it must be tested with residual diagnostics and additional modalities.
2. **Local smoothness (A2).** Within a restricted state region, the latent manifold admits a local smooth metric and differentiable drift approximation. Discrete fate switches, clonal selection, and rare states may violate it.
3. **Directional identifiability (A3).** Directional bias is estimated from time-ordered or perturbation data, not inferred from cross-sectional age labels alone.
4. **Regulatory graph reliability (A4).** The regulatory relations used to define sheaf maps are sufficiently supported by perturbation, chromatin-contact, or carefully qualified evidence. Correlation alone is not causal dependence.
5. **Control observability and reachability (A5).** The state variables needed for constraints are observable at the relevant update cadence, and the intended target is reachable under the available perturbations. Failure of reachability must be reported, not hidden by a large control signal.
6. **Safety surrogate validity (A6).** The identity and risk surrogates correlate with independent functional and pathological endpoints. They cannot replace direct safety measurement.

The framework is not valid merely because an optimizer converges. The assumptions form a checklist against which the data and the model must be audited. When an assumption fails, the appropriate result is model restriction, uncertainty inflation, or rejection of the relevant component.

4 Observables, state representation, and identifiability

The state vector should be constructed around measurable features rather than metaphors. A candidate feature stack includes: (i) cell-type-specific expression modules and transcription-factor activity estimates; (ii) chromatin accessibility and methylation summaries; (iii) 3D contact features such as compartment scores or reproducible domain-level statistics; (iv) chromatin-protein localization or condensate descriptors where valid imaging exists; (v) metabolic context including NAD^+/NADH-related measures and mitochondrial function; (vi) independent lineage-specific functional assays; and (vii) adverse-state indicators such as sustained cell-cycle re-entry, DNA-damage responses, loss of lineage markers, apoptosis, and transformation-associated phenotypes. Which measurements are required depends on the biological system.

Not every experiment can measure every modality longitudinally in the same cell. Multi-omic assays often consume the cell, while repeated imaging can perturb the system and may only observe proxies. Consequently, a practical architecture may combine matched populations, lineage tracing, repeated live-cell measurements, and time-staggered destructive assays. The mapping between measurements and latent state must include uncertainty and technical batch effects.

4.1 Separating chronological age, molecular age, and function

Three targets must remain distinct:

- chronological age is an experimental covariate or elapsed time;

- molecular age is a score trained to correlate with age or a reference molecular state;
- functional rejuvenation is improvement or restoration of a relevant phenotype, such as contractility, barrier function, electrophysiological fidelity, stress tolerance, or tissue repair.

A lower predicted epigenetic age can occur without functional improvement; an age score can also change due to cell-composition or assay-distribution shifts. The primary endpoint in any real experiment must therefore be a prespecified functionally meaningful outcome or a hierarchy of molecular and functional outcomes with controlled multiplicity. CST does not identify a latent age coordinate with literal organismal youth.

4.2 Direction is not causation

RNA velocity and temporal optimal transport can supply directional hypotheses, but they depend on sampling, kinetic assumptions, and the observed cell-state distribution. Cross-sectional datasets do not directly identify within-cell trajectories. A vector field fitted from observational single-cell measurements can be confounded by selection, cell death, proliferation, batch differences, and missing states. Perturbation data and repeated time points are required to attribute a transition to a candidate mechanism. In CST, directional geometry will be learned only where there is sufficient evidence for an oriented transition model; otherwise, the Riemannian baseline should be used.

4.3 Identifiability questions

The following parameters may be confounded: metric curvature and latent-coordinate distortion; directional drift and uneven sampling; sheaf maps and graph-edge uncertainty; dissipation and unobserved driving; and intervention effectiveness and measurement delay. Identifiability will be evaluated by simulation-based parameter recovery, posterior or profile-likelihood diagnostics, intervention contrasts, and held-out prediction. If multiple parameter sets yield indistinguishable observables but divergent control recommendations, the controller is not identified well enough for intervention ranking.

5 Directional state geometry: a cautious Randers-Finsler construction

A Riemannian metric assigns a local quadratic cost to infinitesimal state displacements. A Finsler metric permits a more general direction-dependent norm. A Randers form is

$$F(x, v) = \sqrt{g_{ij}(x)v^i v^j} + \beta_i(x)v^i, \tag{2}$$

where $g(x)$ is positive definite and $\beta(x)$ is a covector field. The standard condition

$$\|\beta(x)\|_g := \sqrt{g^{ij}(x)\beta_i(x)\beta_j(x)} < 1 \tag{3}$$

ensures that F is positive for nonzero velocities and gives a strongly convex Randers structure under the usual regularity assumptions [19, 20]. Units must be handled consistently: the norm of the covector must be dimensionless relative to the metric scaling. In practice, the state variables and time need explicit normalization, or scale parameters must be included.

Proposition 5.1 (Directional asymmetry of a Randers cost)**.** *For any x and velocity v for which*

Equation (2) is defined,

$$F(x, v) - F(x, -v) = 2\beta_i(x)v^i. \tag{4}$$

Proof. The quadratic term is even in velocity because $g_{ij}v^iv^j = g_{ij}(-v^i)(-v^j)$. The linear term changes sign. Subtracting gives twice the linear term. □

This proposition establishes an algebraic asymmetry; it does not establish that aging is a geodesic or that a fitted β is an entropy-production field. The biological hypothesis is narrower: a directional covector estimated from time-resolved transitions may capture a reproducible difference in transition cost between forward and reverse state changes. It should be compared with a symmetric metric and with a directional diffusion/drift model.

5.1 What is the directional covector?

We define β as an empirically estimated covector field linked to the asymmetry in observed transitions after correcting for sampling and covariates. Estimation might proceed by fitting a local transition model and comparing forward and reverse conditional transition likelihoods, or by minimizing a held-out transition prediction objective under the norm constraint Equation (3). The covector is not set equal to entropy production. Entropy production is typically a scalar rate or trajectory functional requiring physical fluxes and thermodynamic forces; it cannot be identified from transcriptomic expression alone. A physically motivated entropy estimate may contribute to a separate cost term if its units, boundaries, and assumptions are documented.

5.2 Trajectory action

For a path $\gamma : [0, T] \to \mathcal{M}$ define the geometric action

$$\mathcal{A}_F[\gamma] = \int_0^T F(\gamma(t), \dot{\gamma}(t))^2 \, dt. \tag{5}$$

The endpoint target is a distribution, not necessarily a single point. A complete objective can combine geometric path action, target-distribution discrepancy, control effort, and penalties:

$$\begin{aligned}\mathcal{J}[\gamma, u] = d_{F,T}(\gamma(T), \mu_{\text{young},c})^2 + \lambda_F \mathcal{A}_F[\gamma] + \lambda_u \int_0^T \|u(t)\|^2 \, dt \\ + \lambda_I \int_0^T [r_I(\gamma(t)) - \epsilon_I]_+^2 \, dt + \lambda_R \int_0^T [r_R(\gamma(t)) - \epsilon_R]_+^2 \, dt.\end{aligned} \tag{6}$$

Here $d_{F,T}$ is a target discrepancy under the chosen geometry, r_I is the identity-surrogate residual and r_R is the risk-surrogate residual. The nonnegative weights are tuning parameters selected without leaking held-out outcomes. $\mathcal{A}_F$ is a path-regularity cost, not automatically physical work. Actual intervention work, ATP consumption, entropy production, or toxicity must be measured separately when available.

5.3 Interpretation of the asymmetry

If the estimated β is stable, improves held-out transition prediction, and remains directionally consistent across technical and biological replicates, it may help characterize barriers to state reversal. If the estimated directional term is unstable across resampling, changes sign under plausible embeddings, or gives no incremental predictive value, its contribution should be removed. A nonzero

β can also arise from incomplete sampling or nonstationary conditions; directionality alone is not evidence of irreversibility at the microscopic level.

6 A regulatory cellular sheaf as an identity diagnostic

A cellular sheaf assigns vector spaces to cells of a graph or cell complex and linear restriction maps that express how local data are compared on shared structure. In CST, let $\mathcal{G}$ be a finite complex derived from a regulatory graph or hypergraph. For each vertex v, let $\mathcal{F}(v)$ be a vector space of local regulatory features; for each edge or higher-dimensional cell e, let $\mathcal{F}(e)$ be a space of compatibility summaries. A restriction map $\rho_{v\to e} : \mathcal{F}(v) \to \mathcal{F}(e)$ encodes the agreed representation of a local feature on that regulatory relation.

For simplicity, consider a graph-level cellular sheaf. The degree-zero cochain space is

$$C^0(\mathcal{G};\mathcal{F}) = \bigoplus_{v\in V} \mathcal{F}(v), \qquad C^1(\mathcal{G};\mathcal{F}) = \bigoplus_{e\in E} \mathcal{F}(e). \tag{7}$$

The coboundary operator $\delta^0 : C^0 \to C^1$ computes disagreement between restrictions of local sections. The degree-zero sheaf Laplacian is

$$L^0_{\mathcal{F}} = (\delta^0)^*\delta^0, \tag{8}$$

where $*$ is the adjoint under the selected inner products. Then

$$\ker(L^0_{\mathcal{F}}) = \ker(\delta^0) \cong H^0(\mathcal{G};\mathcal{F}), \tag{9}$$

for this finite-dimensional setting. Elements of this kernel are globally consistent sections under the specified local restrictions [18].

6.1 What sheaf consistency does and does not mean

The equation above is a mathematical description of consistency given a chosen graph, feature spaces, maps, and inner products. Biological interpretation is supplied by how those ingredients are constructed and validated. If edges are based on correlations, the sheaf residual measures agreement under correlational constraints, not causal regulatory control. If the graph excludes important enhancers, transcription factors, or feedback loops, a low residual may only indicate that the model omitted disagreements. If the sheaf is rebuilt after treatment, a changing kernel can make apparent invariance vacuous unless the comparison is transported through a documented correspondence.

Most importantly, H^0 **preservation does not prove that a cell cannot become cancerous**. Neoplastic transformation can occur through mutations, genomic instability, altered signaling, epigenetic changes, or other routes not represented in the sheaf. It is invalid to describe a sheaf cohomology invariant as a hard theorem of oncological safety. CST uses the sheaf as one regulatory-consistency diagnostic, to be triangulated with independent lineage markers, cell-cycle behavior, genome stability, morphology, and functional assays.

6.2 Identity reference and admissible set

Let $z = \Phi(x)$ map the latent state into the cochain representation. A reference sheaf and reference section set $\mathcal{Q}_c$ are constructed for lineage c from multiple healthy reference donors and, where

appropriate, known functional states. Define

$$r_I(x) = \underbrace{\|\delta^0\Phi(x)\|_W^2}_{\text{regulatory inconsistency}} + \lambda_m \underbrace{d_m(\Phi(x), \mathcal{Q}_c)^2}_{\text{distance to reference identity}} + \lambda_f \underbrace{d_f(q(x), \mathcal{Q}_c^{\text{func}})^2}_{\text{functional deviation}}, \tag{10}$$

where W is a declared positive-semidefinite weighting matrix and the last two terms compare molecular markers and independent function with a lineage-specific reference. The admissible identity region is

$$\mathcal{I}_\varepsilon = \{x : r_I(x) \le \epsilon_I\}. \tag{11}$$

This is more defensible than requiring a raw coordinate vector to lie in the orthogonal complement of a fixed harmonic subspace. It also acknowledges that healthy states vary and that perfect preservation of every baseline coordinate would prevent legitimate adaptation.

6.3 A limitation of hard projection

Projection onto $\ker L_{\mathcal{F}}^0$ can be useful as a regularizer, but a hard projection can suppress biologically meaningful responses or amplify errors from a misspecified graph. It can also be discontinuous if the estimated rank of L changes with noise or treatment. We therefore recommend a soft residual penalty and a sensitivity analysis over graph construction, edge confidence, and regularization. A hard invariant should only be introduced when external evidence supports it and the chosen state representation demonstrates stable, meaningful equivalence across conditions.

7 Port-Hamiltonian-inspired cell dynamics

Port-Hamiltonian systems decompose dynamics into interconnection, dissipation, and external input. For a differentiable scalar function $H_\theta(x)$, define

$$\dot{x} = (J_\theta(x) - R_\theta(x))\nabla H_\theta(x) + B_\theta(x)u + d_\theta(x, z) + \xi(t), \tag{12}$$

where $J_\theta(x) = -J_\theta(x)^\top$, $R_\theta(x) \succeq 0$, B_θ is an input matrix, d_θ represents measured context or unmodeled drift, and ξ is process noise. With output $y = B_\theta^\top \nabla H_\theta$, in a deterministic time-invariant model with $d = 0$ and no noise,

$$\dot{H}_\theta = -\nabla H_\theta^\top R_\theta \nabla H_\theta + y^\top u. \tag{13}$$

The skew-symmetric interconnection contributes no net change to H_θ because $v^\top J v = 0$ for any real vector v; the R term is nonpositive; the input port may add or remove energy relative to this learned function.

The model is useful because it constrains the learned dynamics instead of allowing an arbitrary neural vector field. But a learned H_θ is not automatically measured Gibbs free energy, ATP cost, or cellular enthalpy. Unless the model is grounded in physically calibrated state variables and energetic measurements, H_θ should be described as a learned storage/Lyapunov-like function. Likewise, J and R are a parameterization of an inductive bias; fitting them does not establish that mitochondria are a conservative oscillator and chromatin drift is a purely dissipative process.

7.1 Metabolic and phase variables as candidate features

A candidate state may include oscillator phases ϑ_k for metabolic or transcriptional signals. A Kuramoto-inspired summary is

$$Z(t) = \frac{1}{N}\sum_{k=1}^{N} w_k e^{i\vartheta_k(t)}, \qquad r_\phi(t) = 1 - |Z(t)|, \tag{14}$$

where weights w_k and phase definitions are specified before testing. This is only meaningful when time series support stable phase estimation; single snapshots cannot recover oscillation phase. The synchrony statistic is not a universal aging measure. It should be evaluated as an additional covariate and tested for incremental predictive value over metabolite levels, circadian phase, mitochondrial function, cell cycle, and environmental context.

7.2 Model classes and baselines

We recommend three nested model classes: (1) a conventional state-space or neural ODE without structured energy constraints; (2) the port-Hamiltonian-inspired model with input-output structure; and (3) the same model plus candidate phase features. All should share comparable data splits, parameter budgets where possible, uncertainty calibration, and missing-modality handling. A structured model should be retained only if it improves out-of-sample prediction, calibration, or intervention-effect estimation, not because it looks more physically sophisticated.

8 Counterdiabatic-style control with reachability checks

Suppose the learned uncontrolled drift is

$$f_\theta(x, z) = (J_\theta(x) - R_\theta(x))\nabla H_\theta(x) + d_\theta(x, z). \tag{15}$$

Given a proposed reference path $\gamma_*(t)$, the instantaneous residual velocity is $r(t) = \dot{\gamma}_*(t) - f_\theta(\gamma_*(t), z_t)$. A formal exact input would solve $B_\theta(x)u = r$. This is feasible only if $r \in \mathrm{Range}(B_\theta(x))$. A Moore-Penrose pseudoinverse returns a least-squares solution when exact matching is impossible; it does not guarantee safety, boundedness, or feasibility.

We therefore define an approximate control step

$$u^*(t) = \arg\min_{u\in\mathcal{U}} \left\{ \|B_\theta u - r\|^2_{W_u} + \lambda_u \|u\|_2^2 \right\} \tag{16}$$

subject to the state dynamics, input amplitude and slew constraints, identity set $x_t \in \mathcal{I}_\varepsilon$, risk guards, and an uncertainty margin. The allowed input set $\mathcal{U}$ represents real candidate perturbations only after their directions, delivery mechanism, dose range, and observed effects are characterized. It should not mix transcript-factor pulses, bioelectric stimulation, and small molecules as though they were interchangeable controls.

8.1 Safety constraints are independent of topology

Let $h_k(x) \le 0$ denote empirically defined guard functions for such outcomes as sustained proliferation, a lineage-specific function score, DNA-damage responses, or transformation-related phenotypes. The set

$$\mathcal{S}_{\text{safe}} = \{x : r_I(x) \le \epsilon_I,\ h_k(x) \le 0 \text{ for all } k\} \tag{17}$$

can be used in model-predictive control or chance-constrained optimization. In a stochastic model, a chance constraint could be written

$$\Pr\left(\sup_{t\in[0,T]} h_k(x_t) > 0\right) \le \alpha_k. \tag{18}$$

A guarantee is only as credible as the probability model, calibration, coverage of rare events, and validation data. Without a sufficient biological safety dataset, these are design targets rather than certified bounds. No in-silico model can establish absence of tumorigenicity.

8.2 The control target may be infeasible

Identity preservation, target closeness, and low control energy can be mutually incompatible. The framework must report the attainable Pareto frontier rather than return a single overconfident "optimal" intervention. A solution should be declared infeasible if no allowed control reaches a prespecified target while satisfying constraints under uncertainty. The target can then be relaxed only through a transparent scientific decision and a preregistered revision, not by silent tuning after seeing outcomes.

9 Joint optimization problem

Let $u(\cdot)$ be an admissible control and let $x(\cdot)$ satisfy the model. A risk-aware objective is

$$\begin{aligned}\min_{u(\cdot)} \quad \mathbb{E}\Bigg[& D_{\text{target}}(x_T, \mu_{\text{young},c}) + \lambda_F \int_0^T F(x_t, \dot{x}_t)^2\, dt \\ & + \lambda_u \int_0^T \|u_t\|^2 dt + \lambda_I \int_0^T [r_I(x_t) - \epsilon_I]_+^2\, dt \\ & + \lambda_R \int_0^T [r_R(x_t) - \epsilon_R]_+^2\, dt + \lambda_S\, \mathcal{R}_{\text{uncertainty}}(u) \Bigg]\end{aligned} \tag{19}$$

subject to Equation (12), $u_t \in \mathcal{U}$, initial state distribution $x_0 \sim \mu_{\text{aged},c}$, identity constraints, and risk guardrails. D_{target} must not be chosen solely because it is easy for the model to optimize; at least one endpoint must remain independent of the state representation used for control.

A robust version optimizes against a set of plausible dynamics Θ or worst-case risk:

$$\min_u \max_{\theta\in\Theta} \mathcal{J}_\theta(u), \qquad x_t \in \mathcal{S}_{\text{safe},\theta} \text{ for plausible } \theta. \tag{20}$$

This version may be computationally expensive and can produce conservative controls. A practical study should begin with small systems and compare nominal model predictive control, constrained linear-quadratic control where applicable, a neural ODE policy, and a simple pulsed baseline. The counterdiabatic component has value only if it improves the risk-benefit frontier under matched targets and comparable control resources.

9.1 Proposed algorithm

The algorithm is modular by design. The state estimator produces a posterior distribution rather than one exact x_t. The geometry module estimates the metric and directional covector. The sheaf module produces a consistency residual and its uncertainty. The dynamics module predicts states under candidate inputs. The optimizer selects a bounded action or declares infeasibility. During the

control proposal, no action is applied to a biological system until independent experimentation has calibrated the relevant transitions and guard functions.

Algorithm 1 CST inference and candidate-control loop (research blueprint)

Require: Matched time-resolved data Y, lineage label c, reference distributions, candidate input set $\mathcal{U}$

Ensure: Predictive model, risk/identity diagnostics, and either a candidate policy or an infeasibility report

1: Fit a measurement model and latent state estimator with donor/study-aware validation.
2: Estimate the information metric g and test a symmetric baseline before fitting directional β.
3: Retain β only if regularity, directionality, and held-out prediction checks pass.
4: Construct a documented regulatory complex $\mathcal{G}$ and uncertain restriction maps; compute $r_I(x)$.
5: Fit unconstrained and port-Hamiltonian-inspired dynamics with identical data splits.
6: Calibrate risk surrogates using independent functional and pathological observations where available.
7: Propose a reference path from the aged-state distribution toward the lineage-matched target distribution.
8: Solve the bounded least-squares control problem with identity and risk guards.
9: **if** reachability or uncertainty constraints fail **then**
10: Return "infeasible / insufficiently identified" and the active failure reason.
11: **else**
12: Return candidate schedule, uncertainty envelope, model version, and preregistered validation tests.
13: **end if**
14: Evaluate only on held-out donors, perturbations, and endpoints; record all negative results and ablations.

10 Synthetic demonstration: methods and results

Data status. This section uses a hand-constructed four-state differential equation and a fixed random seed. No biological data, measured parameter, training atlas, or published single-cell count was used. The numerical experiment exists to verify code execution and to expose the consequences of selected structural assumptions. It is not evidence that CST predicts or safely controls living cells.

10.1 Toy dynamics

The dimensionless state is $x = (a, \ell, r, \phi)$, where a is an abstract age-associated coordinate (lower is labelled the target direction), ℓ is an identity-deviation proxy, r is an oncogenic-stress proxy, and ϕ is a phase-decoherence proxy. These names are labels within the toy system, not biological biomarkers. The dynamics are

$$\dot{a} = k_a + k_{a\phi}\phi - \gamma_a a - \eta_a u, \tag{21}$$

$$\dot{\ell} = -\gamma_\ell \ell + \eta_\ell u^2 - \rho_\ell v, \tag{22}$$

$$\dot{r} = -\gamma_r r + \eta_r u^2 - \rho_r v, \tag{23}$$

$$\dot{\phi} = -\gamma_\phi \phi + \eta_\phi u, \tag{24}$$

where $u \geq 0$ is a generic reprogramming input and $v \geq 0$ is a generic repair input. The quadratic coupling u^2 means high inputs carry larger identity- and risk-proxy costs in this deliberately chosen

example. Nothing in biology compels this exact equation; different models may produce different rankings.

The starting state is $(1.0, 0.02, 0.01, 0.18)$ and the integration interval is $T = 12$ dimensionless time units with step $\Delta t = 0.01$. The base coefficients are $k_a = 0.018$, $k_{a\phi} = 0.040$, $\gamma_a = 0.10$, $\eta_a = 0.24$, $\gamma_\ell = 0.40$, $\eta_\ell = 0.55$, $\rho_\ell = 0.45$, $\gamma_r = 0.38$, $\eta_r = 0.45$, $\rho_r = 0.35$, $\gamma_\phi = 0.20$, and $\eta_\phi = 0.25$. All coefficients are synthetic and unitless. The baseline has no control; the aggressive protocol applies $u = 0.90$ from $t = 0.4$ through 4.4 without repair; the constrained protocol applies $u = 0.62$ in pulses when the identity and risk proxies are below illustrative thresholds and uses a feedback repair input. Details are fully specified in `simulate_model.py`.

10.2 Deterministic demonstration

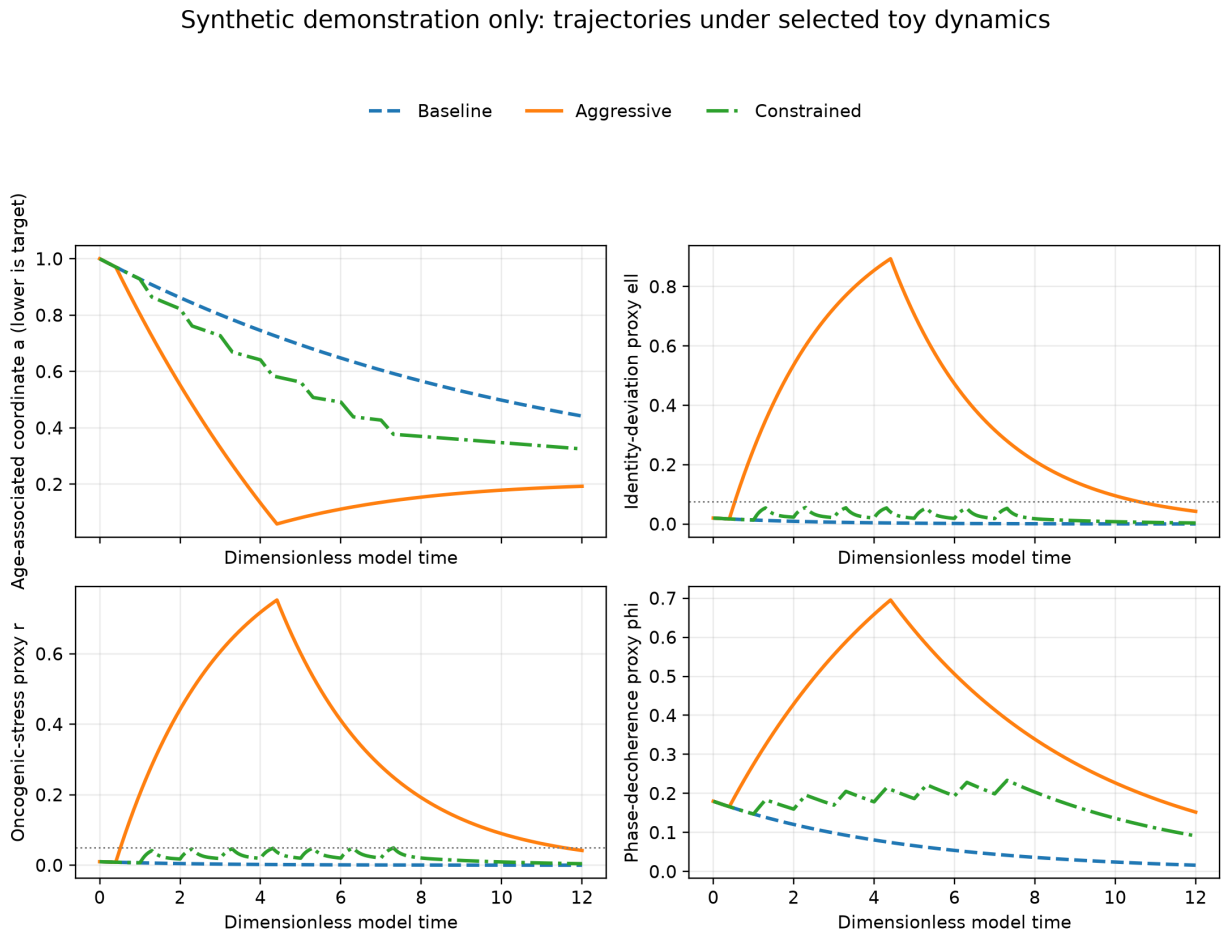


Figure 1: Synthetic trajectories under the selected toy model. The age-associated coordinate is a model label; identity and risk axes are synthetic proxies. The dotted lines are illustrative guardrails, not clinical thresholds.

The aggressive policy lowers the endpoint age-associated coordinate from 1.0 to approximately 0.193, but its maximum identity-deviation proxy reaches 0.894 and its maximum risk proxy reaches 0.753. The constrained pulsed policy ends at approximately 0.325, with maximum identity deviation 0.056 and maximum risk proxy 0.049. The no-control baseline ends at approximately 0.442 with the lowest input effort. These numbers follow from the constructed coefficients. They demonstrate a logical point—endpoint optimization and constrained optimization can choose different trajectories—not a biological effect size.

Table 2: Deterministic outcomes of the toy model (unitless).

Policy	Final a	Peak ℓ	Peak r	Peak ϕ	$\int u^2 dt$	$\int v^2 dt$
Baseline	0.442	0.020	0.010	0.180	0.000	0.000
Aggressive	0.193	0.894	0.753	0.695	3.248	0.000
Constrained pulse	0.325	0.056	0.049	0.234	0.823	0.160

10.3 Parameter/noise stress test

We generated 300 replicates. For each replicate, every base coefficient was independently multiplied by a factor sampled uniformly between 0.8 and 1.2, and independent normalized process noise was added to the state derivatives. This is a deliberately broad but arbitrary robustness check, not a posterior distribution learned from biology. All results use seed 20261014. The experiment checks whether the toy model's qualitative trade-off persists under its chosen parameter perturbations.

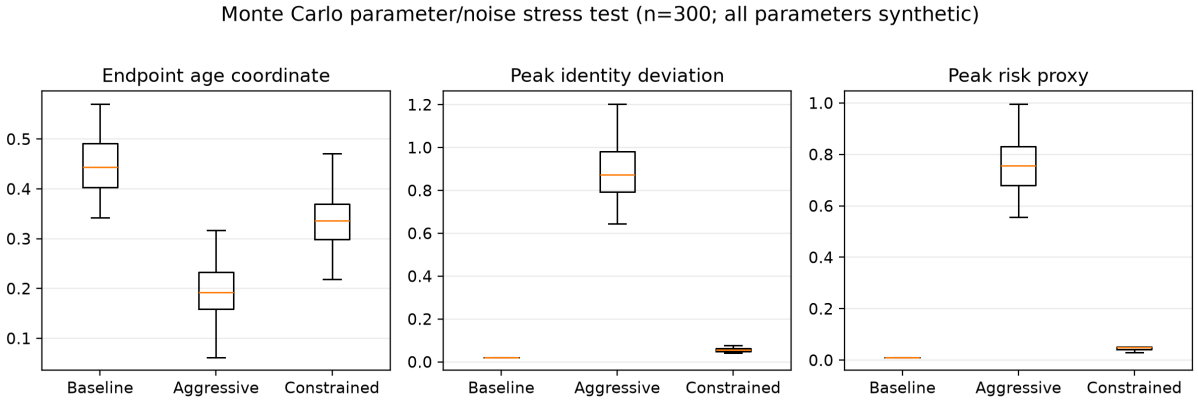


Figure 2: Monte Carlo distributions across 300 synthetic parameter/noise replicates. The boxes do not represent biological uncertainty; they summarize only the stated toy simulation.

Table 3: Monte Carlo medians and 5th–95th percentile ranges across 300 synthetic replicates.

Policy	Quantity	Median [5th, 95th percentile]
Baseline	Final age coordinate	0.443 [0.370, 0.536]
Baseline	Maximum identity deviation	0.020 [0.020, 0.020]
Baseline	Maximum risk proxy	0.010 [0.010, 0.011]
Aggressive	Final age coordinate	0.192 [0.118, 0.277]
Aggressive	Maximum identity deviation	0.874 [0.693, 1.099]
Aggressive	Maximum risk proxy	0.756 [0.623, 0.934]
Constrained pulse	Final age coordinate	0.336 [0.264, 0.417]
Constrained pulse	Maximum identity deviation	0.055 [0.045, 0.073]
Constrained pulse	Maximum risk proxy	0.050 [0.035, 0.051]

10.4 What the synthetic results do and do not establish

They establish that the implementation produces finite trajectories, respects the non-negativity handling for deviation proxies, reproduces a known trade-off under the selected differential equations, and reports a fixed-seed parameter/noise stress test. They do not test the Randers metric, sheaf cohomology, or learned port-Hamiltonian model on real measurements; those modules are presented

formally and remain unimplemented as biological estimators in this initial release. This distinction matters: the toy ODE intentionally embodies an identity/risk cost for aggressive input. Its results cannot support the claim that real reprogramming necessarily causes the same quantitative trade-off or that CST is safer than a biological protocol.

The correct next computational step is not to enlarge this toy model until it appears convincing. It is to validate the state estimator, graph construction, directional drift, and control model separately against observations they did not train on. If the real data fail to support one module, the module must be removed or revised.

11 Empirical programme and data feasibility

The proposed validation should proceed in stages. A dataset inventory must precede model construction. Its purpose is to determine which modalities, cell types, ages, replicates, time points, perturbations, and functional outcomes are actually available together. Combining an aging atlas with a different reprogramming dataset does not create a matched longitudinal cohort; it creates a cross-study transfer problem requiring explicit harmonization and additional uncertainty.

11.1 Stage 0: dataset and assay audit

The Tabula Muris Senis dataset is a reasonable starting point for cell-type-specific age-associated expression signatures across 23 tissues [2]. Its primary modality is not a matched, longitudinal multiome of RNA, chromatin contacts, metabolic phase, and function for each cell. It can support the transcriptomic arm of model development, but it cannot alone validate a Finslerian chromatin manifold or infer mitochondrial phase dynamics. A separate search of NCBI Gene Expression Omnibus, ENCODE, and the 4D Nucleome portal should identify candidate time-resolved reprogramming, accessibility, contact-map, and perturbation datasets. Each candidate should be reviewed for donor replication, lineage specificity, assay comparability, treatment timing, and availability of an independent function or safety endpoint before inclusion.

At manuscript preparation time, no listed biological dataset was downloaded or analyzed for CST. The links and accessions specify a proposed route, not a declaration that these datasets jointly contain every required modality. Before a future analysis, a curator should create a machine-readable manifest that includes persistent accession, file checksum, license, sample-level metadata availability, biological replicate, cell type, time/age, treatment, modality, missingness, and exclusion reasons.

11.2 Stage 1: cross-sectional reference-state estimation

The first analysis would estimate lineage-specific reference distributions and age-associated molecular axes using donor-aware splitting. Cells from one donor, animal, plate, or experiment must not be split between training and test sets in ways that create leakage. The model should distinguish within-cell-type age shifts from cell-composition shifts. Molecular-age scores can be used as secondary endpoints, but the target distribution and metric should not be tuned until they produce a favorable intervention result.

11.3 Stage 2: directional transition estimation

Time-resolved or perturbation datasets are needed to estimate β. RNA velocity is one candidate method but has assumptions that require diagnostic checks [15, 16]. Alternative estimates can be

Table 4: Candidate public-data resources and what they can realistically support. Availability must be re-audited before a future empirical study.

Resource	Primary role	Limitations for CST
Tabula Muris Senis / GSE132042	Mouse tissue and cell-type expression across lifespan [2]	Cross-sectional single-cell expression is not same-cell longitudinal dynamics; not a complete matched chromatin/metabolic/function panel.
NCBI GEO	Search for time-resolved reprogramming and aging transcriptomics	Every series needs treatment, replicate, batch, and endpoint audit; many studies are small or incomparable.
ENCODE	Regulatory annotations, accessibility, selected chromatin assays	Heterogeneous cell types and conditions; annotations are evidence sources, not automatically a lineage-specific causal graph.
4D Nucleome	3D genome organization and contact-map protocols/data	Data must match cell identity, age/treatment, and sampling design; chromatin contacts alone do not identify transcriptional dynamics.
Published partial-reprogramming datasets	Candidate intervention trajectories and endpoint benchmarks [3, 4, 5]	Published papers may not provide all raw data or same-cell multimodal time courses; matched function and safety data may be limited.

based on optimal transport across known time points [14], lineage tracing, or repeated live-cell measurements. All approaches should compare directional and symmetric models, evaluate calibration of transition probabilities, and check stability against cell-cycle composition, proliferation/death, and sampling imbalance.

11.4 Stage 3: regulatory-sheaf construction

The graph must record how each edge was supported: perturbation evidence, curated regulatory interaction, high-confidence contact, or lower-confidence correlation. Edge weights should reflect uncertainty rather than convert every correlation into a hard restriction map. Sensitivity analysis should vary thresholds and graph density. Candidate outcomes include sheaf consistency residual, selected spectral quantities, and rank stability. Those quantities must then be tested for incremental association with independent lineage markers and function, beyond simpler graph smoothness and marker-based baselines.

11.5 Stage 4: control benchmarking

Only after transition dynamics and safety surrogates are calibrated should control be evaluated. At minimum, compare: (i) no-control and pulse baselines; (ii) Euclidean endpoint matching; (iii) Riemannian path control; (iv) Finsler directional cost without sheaf penalty; (v) sheaf penalty without directional cost; (vi) conventional model predictive control; (vii) a generic neural-ODE controller; and (viii) full CST. Each ablation isolates a claimed contribution. Inputs must be limited to interventions that are observable, delivered with specified dose/timing, and supported by experimental response data.

12 Pre-registered hypotheses, metrics, and decision rules

The following hypotheses are proposals for a future preregistered study. They are not results from the present simulation.

12.1 H1: directional geometry adds transition information

Hypothesis. The fitted directional covector β improves held-out next-state or transition-probability prediction relative to a symmetric metric after controlling for time, lineage, donor, cell cycle, and batch. **Primary metric.** Test-set negative log-likelihood or a proper scoring rule on donor- and study-disjoint data, with uncertainty intervals from resampling biological replicates. **Falsification.** No reliable improvement after calibration, or signs/magnitudes of β vary arbitrarily across reasonable feature maps and resamples. The Finsler term is removed if falsified.

12.2 H2: sheaf residual provides independent identity information

Hypothesis. A regulatory-sheaf residual predicts independently measured lineage deviation or function beyond standard marker-gene distances, cell identity classifiers, and graph smoothness baselines. **Primary metric.** Incremental held-out predictive log score or predeclared AUROC/AUPRC where the outcome is categorical, with donor-level confidence intervals. **Falsification.** The residual adds no predictive information, is dominated by marker genes, or changes substantially under small plausible graph perturbations. A simple marker or network model replaces the sheaf component.

12.3 H3: phase-related features add conditional predictive value

Hypothesis. Time-resolved metabolic or transcriptional phase descriptors improve prediction of subsequent chromatin-state change after conditioning on age, treatment, cell cycle, and metabolite abundance. **Falsification.** No out-of-sample gain, or the estimated phase signal is a sampling artifact. If oscillatory variables are not observed repeatedly, H3 is not tested rather than inferred from cross-sectional data.

12.4 H4: structured control improves the trade-off

Hypothesis. At matched independent molecular/functional improvement, a constrained controller produces lower identity drift, lower risk-proxy excursion, or lower measured control burden than an unconstrained controller under comparable input limits. **Primary metric.** A predeclared Pareto comparison over target distance, functional outcome, lineage deviation, safety, and input burden. **Falsification.** CST fails to improve the frontier over simpler MPC or pulsed baselines, or any apparent gain disappears in held-out donors or perturbations.

12.5 Statistical plan

The independent unit is the biological replicate, not the individual cell. Use donor/animal-level stratified splitting and nested cross-validation for hyperparameter selection. Report calibration, uncertainty intervals, missingness, failed optimization runs, and the proportion of target-control problems declared infeasible. For multiple primary hypotheses, apply a prespecified multiplicity correction or a hierarchical testing order. Use negative controls and label permutations to detect leakage. Evaluate cell-type classifiers and lineage-marker panels on data not used in feature selection.

If a perturbation dataset has too few biological replicates to provide meaningful confidence bounds, treat it as exploratory and do not promote the result as confirmatory.

13 Falsification ladder and model-ablation discipline

A high-complexity theory can appear successful because each component offers another opportunity to fit noise. CST is therefore governed by a staged falsification ladder:

1. **Measurement failure:** if modalities cannot be aligned or sample provenance is insufficient, do not fit a joint manifold.
2. **Geometry failure:** if the metric is unstable or out-of-sample directionality is not improved, retain a simpler symmetric metric.
3. **Sheaf failure:** if the sheaf score adds no independent identity information or is graph-construction sensitive, remove the topology term.
4. **Dynamics failure:** if the port-Hamiltonian parameterization is no better calibrated than simpler models, use the simpler model.
5. **Reachability failure:** if the desired velocity lies outside the feasible actuation range, report infeasibility rather than increasing control magnitude or relaxing safety silently.
6. **Safety-surrogate failure:** if identity or risk scores do not correlate with external functional/pathological endpoints, do not use them as control constraints.
7. **Translation failure:** if molecular markers improve without independent function, the result is molecular-state modulation, not demonstrated rejuvenation.

A component survives only if it makes a reproducible difference against a prespecified baseline. A framework can be useful even if only part of it survives; it should not be protected from falsification by adding more latent variables after an unfavorable result.

14 Threats to validity and common failure modes

14.1 Latent-space geometry can be an artifact

A nonlinear embedding can make Euclidean distances look curved or asymmetric even if the underlying transition structure is not. Metrics should be estimated in a probabilistic model where feasible, checked across embeddings, and evaluated in coordinate-independent terms. When feature scaling changes $\|\beta\|_g$, the Randers condition can be satisfied or violated by a rescaling if units are not carefully defined. Preprocessing is part of the model and must be frozen during testing.

14.2 Age is a confounder, not a trajectory label

Cell populations observed at young and old ages are not necessarily ancestor-descendant samples of each other. Cells may be selectively lost, recruited, proliferate, or change cell-type composition. A path between two means is not proof that a given cell can traverse that path under intervention. The study design should use temporal sampling, perturbations, and/or lineage information to distinguish transition from selection.

14.3 Cohomology depends on the model complex

A topological invariant is invariant relative to specific mathematical structure and transformations. It does not preserve biological identity independently of representation. If nodes, edges, restriction maps, or reference kernels are inferred from noisy data, topological summaries may change because of thresholding rather than biology. The model should quantify uncertainty over complexes and compare the cohomology score with standard graph and marker baselines.

14.4 Thermodynamic language requires thermodynamic evidence

The action $\mathcal{A}_F$, learned Hamiltonian H_θ, control energy $\int \|u\|^2 dt$, metabolic ATP use, and entropy production are distinct quantities. They should never share the same label without an explicit calibrated mapping. Cell state is open to nutrient supply, signaling, cell division, and environment; any local energy balance must state the boundary of the modeled system. A formal decrease in a neural network's storage function is not evidence of lower entropy production in the biological system.

14.5 Rare but catastrophic failure is hard to bound

A controller optimized under average loss may ignore rare dedifferentiation, clonal expansion, or oncogenic states. A finite dataset may contain no examples of an event that still has nonzero risk. Chance constraints are only as good as the tails represented in the model. For early-stage studies, the framework should be confined to in-silico and ex-vivo hypothesis prioritization, with independent oversight and established assays; it must not be used to recommend untested human interventions.

15 Safety, ethics, and translational boundary

Cellular reprogramming has potential for regenerative medicine but also raises questions about lineage stability, abnormal proliferation, genomic integrity, delivery, tissue mosaicism, durability, and long-term observation. The literature explicitly describes safety concerns and the need for temporal control in partial cellular reprogramming [7, 6]. A geometric model cannot eliminate these issues by definition.

CST is a research framework, not a treatment protocol. It proposes no human dose, delivery vector, target gene list, clinical schedule, or recommendation to apply OSKM or any substitute. A translational programme would require model validation, appropriate cell-type-specific experimental work, independent pathology, dose-response characterization, replication in relevant animal models where justified, and regulatory/ethical review. The threshold constraints in the model are not established clinical thresholds.

Data governance also matters. Public single-cell datasets can contain sensitive biological information and licensing requirements. A future data release should retain provenance, cite accessions, distinguish derived features from source files, and document exclusions and preprocessing. AI systems should not be represented as having independently conducted experiments or verified claims beyond their available tools. Human investigators retain responsibility for scientific and ethical decisions.

16 Discussion: the strongest defensible interpretation

The intellectual ambition of CST is to make identity-aware rejuvenation a formal control problem rather than a task of moving a molecular age score backward. The mathematical components make different demands on evidence. Information geometry asks whether local statistical distances are meaningful. The directional covector asks whether there is a reproducible asymmetry in observed transition processes. The sheaf model asks whether local regulatory features agree under a documented set of relations. The port-Hamiltonian model asks whether an energy-accounting inductive bias improves dynamical prediction. Counterdiabatic-style control asks whether an auxiliary input can track a feasible trajectory while avoiding unacceptable excursions. None of these questions can be answered by the elegance of the equation alone.

The asymmetry term is potentially useful because epigenetic transitions can be path-dependent and history-dependent. But a state transition inferred from data may be asymmetric because of sampling design, proliferation, cell selection, or unmeasured inputs. The framework explicitly leaves open a symmetric outcome. The sheaf construction is potentially useful because regulatory features have local constraints and global consistency requirements; however, a low sheaf residual is not itself a validated biological identity score. Its scientific meaning depends on edge evidence and external prediction.

The port-Hamiltonian structure imposes a disciplined accounting identity that may regularize learning. The model's learned storage function, however, should not be called thermodynamic work unless calibrated to physical units and boundaries. Likewise, counterdiabatic control is a design inspiration. In a living cell, the actuators may be weak, slow, nonlocal, stochastic, or unavailable. The correct outcome of the controller may often be "not reachable with current perturbations." That is a useful scientific conclusion, not an algorithmic failure.

The synthetic experiment illustrates the logic of constrained optimization: if a constructed model makes high input improve one coordinate while quadratically worsening identity and risk proxies, an aggressive controller will trade away those proxies for endpoint gain. A constrained pulse controller gives a different point on the trade-off frontier. That observation is nearly tautological given the selected equations. Its value is software verification and an explicit example for testing pipeline behavior. It provides no estimate of the real biological trade-off. A future data-grounded model may show that gentle pulses are not superior, that the proposed risk proxy is inadequate, or that the target state is not reachable while maintaining function.

We also reject the assumption that rejuvenation must reverse time or return a cell to an exact youthful transcriptome. A cell may benefit from restoration of specific functions while retaining disease history or age-associated features that are adaptive. An optimal state could be functional, lineage-stable, and safer without minimizing every molecular age statistic. The reference distribution must be defined scientifically for the biological context rather than chosen as the youngest point in a dataset.

16.1 A practical route to scientific value

The immediate useful outcome is an experimental-computational protocol: first measure a narrow lineage and intervention context; then verify the direction field, then quantify identity using independent endpoints, and only then test constrained control. A small validated module can be more informative than a large unvalidated architecture. For example, if a sheaf residual improves prediction of cardiomyocyte electrophysiological fidelity after partial reprogramming while controlling for standard marker panels, that would justify further study of the topology term. If it does not, a

simpler marker-based constraint is preferable.

CST's potential value is therefore conditional and testable. It offers a language for separating trajectory geometry, regulatory agreement, control effort, and function while making it harder to confuse a successful simulation with biological discovery. Such separation is essential if AI-assisted modeling is to help rather than overstate longevity science.

17 Conclusion

We have defined Counterdiabatic Sheaf Transport as a candidate framework for identity-constrained, energy-aware modeling of partial cellular rejuvenation. The formulation combines an information-geometric state space, a regular Randers-Finsler path cost, a regulatory cellular-sheaf residual, port-Hamiltonian-inspired dynamics, and reachability-aware constrained control. We show the mathematical conditions required by the model, explain why the directional covector cannot simply be equated with entropy production, and demonstrate why neither cohomology nor a learned storage function guarantees biological safety.

A synthetic four-state simulation and a 300-replicate parameter/noise stress test confirm the implementation and illustrate how endpoint optimization can differ from safety-constrained control in a toy model. They do not validate the model against living cells. No biological dataset or experiment was analyzed. The next test is a staged, donor-aware empirical study comparing each CST component with simpler baselines and independent function and safety endpoints. If the directional term, sheaf score, or structured controller does not improve held-out performance, that component should be removed.

The bold claim worth testing is not that topology makes cancer impossible, or that one geometry explains aging. It is that a careful mathematical separation of state change, lineage coherence, energy accounting, control reachability, and independently measured function could improve how rejuvenation hypotheses are designed and falsified. The scientific merit of CST depends entirely on whether that separation yields better experiments and reproducible evidence.

A Mathematical notes and conditions

A.1 Randers positivity and directionality

For a positive-definite matrix g, denote $\alpha(v) = \sqrt{g(v,v)}$ and $\beta(v) = \beta_i v^i$. By Cauchy-Schwarz under the g pairing,

$$|\beta(v)| \leq \|\beta\|_g \, \alpha(v). \tag{25}$$

If $\|\beta\|_g < 1$, then for $v \neq 0$,

$$F(v) = \alpha(v) + \beta(v) \geq (1 - \|\beta\|_g)\alpha(v) > 0. \tag{26}$$

This establishes positivity. The standard Randers regularity result additionally requires the metric and covector fields to meet the stated smoothness assumptions. These conditions should be checked numerically at every evaluated state. If a fitted covector norm approaches one, model uncertainty is large, or the metric becomes ill-conditioned, the estimator should shrink β toward zero or revert to a symmetric baseline.

The directional gap $F(v) - F(-v) = 2\beta(v)$ is an algebraic identity. It does not prove that the physical process is irreversible: the asymmetry may reflect biased sampling, an external field, or a coarse-grained state description. Physical entropy production is a distinct measurable quantity requiring a thermodynamic model.

A.2 Sheaf consistency residual and kernel

Let $\delta : C^0 \to C^1$ be a linear coboundary operator on finite-dimensional inner-product spaces. Then $L^0 = \delta^* \delta$ is positive semidefinite because

$$\langle z, L^0 z \rangle = \langle z, \delta^* \delta z \rangle = \|\delta z\|^2 \geq 0. \tag{27}$$

If $L^0 z = 0$, then $0 = \langle z, L^0 z \rangle = \|\delta z\|^2$, hence $\delta z = 0$. Conversely, if $\delta z = 0$, then $L^0 z = 0$. Therefore $\ker L^0 = \ker \delta$, the space of global sections in this discrete sheaf model. This statement is mathematical and exact for the chosen sheaf. Its relation to lineage identity is an empirical modeling claim, not a consequence of the kernel identity.

A.3 Port-Hamiltonian storage balance

For deterministic dynamics $\dot{x} = (J - R)\nabla H + Bu + d$ with $J = -J^\top$ and $R \succeq 0$, the chain rule gives

$$\begin{aligned} \dot{H} &= \nabla H^\top \dot{x} && (28)\\ &= \nabla H^\top J \nabla H - \nabla H^\top R \nabla H + \nabla H^\top B u + \nabla H^\top d && (29)\\ &= -\nabla H^\top R \nabla H + y^\top u + \nabla H^\top d, && (30) \end{aligned}$$

where $y = B^\top \nabla H$. The first term vanishes because a scalar $w^\top J w$ equals its own negative and therefore is zero. For $d = 0$, this yields Equation (13). With explicit time dependence in H, process noise, state estimation error, or stochastic calculus, additional terms appear. Thus the compact balance is not directly applicable to an arbitrary neural ODE with noise without the corresponding derivation.

A.4 Exact tracking feasibility

For instantaneous exact tracking, the residual $r = \dot{\gamma}_* - f_\theta(x, z)$ must belong to the column space of $B_\theta(x)$. If it does not, no input can produce the requested velocity at that state, regardless of optimization quality. The regularized least-squares solution in Equation (16) minimizes residual mismatch plus effort. Its solution must still respect box constraints, rate limits, uncertainty, and state-dependent safety constraints. A pseudoinverse alone does not supply any of these guarantees.

B Model variables and units

Table 5: Notation and interpretation.

Symbol	Definition	Status / requirement
x	Latent cell-state representation	Estimated with uncertainty; not directly observed.
Y	Multimodal measurements	Assay-specific counts, images, contact features, metabolites, function.
$\mathcal{M}$	Local latent-state manifold	Approximation requiring stable local geometry.
g_{ij}	Information/Riemannian metric	Estimated; must be regularized and tested across embeddings.
β_i	Directional covector	Hypothesis estimated from temporal/perturbation evidence; not entropy production by definition.
$F(x, v)$	Randers-Finsler path norm	Valid only if g positive definite and $\|\beta\|_g < 1$.
$\mathcal{G}$	Regulatory graph or cell complex	Must track edge evidence and uncertainty.
$\mathcal{F}$	Cellular sheaf over $\mathcal{G}$	Model of local regulatory representations/compatibility.
δ^0	Degree-zero coboundary	Maps local sections to compatibility residuals.
$L^0_{\mathcal{F}}$	Sheaf Laplacian $\delta^{0*}\delta^0$	Positive semidefinite for declared inner products.
r_I	Identity surrogate residual	Requires external validation against lineage/function.
H_θ	Learned storage/Hamiltonian function	Not physical free energy unless calibrated.
J_θ	Skew-symmetric interconnection matrix	Structural inductive bias.
R_θ	Dissipation matrix	Must be constrained positive semidefinite.
B_θ	Control input matrix	Encodes actuation range and limitations.
u	Candidate rejuvenation input	Only meaningful for a specified experimental intervention.
v	Illustrative repair input in toy simulation	Synthetic-only variable; not a specific therapy.

Symbol	Definition	Status / requirement
r_R	Risk surrogate	Requires validation against independent safety endpoints.
$\mu_{\text{young},c}$	Lineage-specific reference distribution	Frozen before evaluation; not necessarily a single ideal state.

C Reproducibility and reporting checklist

The accompanying script `simulate_model.py` is self-contained apart from NumPy and Matplotlib. It sets the seed to 20261014 and records all base coefficients in `assets/synthetic_summary.json`. To reproduce the numerical demonstration, run:

```
python simulate_model.py
```

The script regenerates figures and summary output. The output should be labeled synthetic in every downstream report.

A future biological study should add the following before fitting a controller:

1. machine-readable dataset and sample manifest with checksums, licenses, and exclusions;
2. frozen preprocessing and reference-state definition;
3. donor/animal/study-level train-validation-test splits;
4. an independently measured lineage function endpoint;
5. a prespecified adverse-state panel and pathological follow-up appropriate to the experiment;
6. calibration and uncertainty coverage for state and transition predictions;
7. ablation of the Finsler term, sheaf term, phase variables, and port-Hamiltonian structure;
8. a report of infeasible target paths and failed runs;
9. versioned code, parameter configs, seeds, and environment lockfile;
10. clear separation between model-predicted benefits and measured experimental outcomes.

D Limitations summary

This preprint does not contain a biological analysis, new empirical measurements, a trained cell-state controller, or an intervention recommendation. The formal and simulation components support an executable research specification but do not establish molecular causality. The sheaf model is a proposed regularizer; the directional metric is a proposed estimator family; the port-Hamiltonian parameterization is a candidate model class; and the controller is an optimization blueprint. Existing work already covers partial reprogramming, information-geometric cell states, regulatory networks, phase separation, and energy-structured learning. The novelty and usefulness of their integration remain open. The work should be treated as a hypothesis-bearing methods preprint and subjected to expert review before any empirical or translational use.

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